Print out the following set of Monohybrid and Dihybrid problems. Complete the punnett squares and answer the question in each problem.

Question Set A.

Albinism is a condition in humans in which they are unable to produce the pigment melanin. People with this condition take on the color of the blood flowing through their capillaries. This occurs when the dominant normal allele (N) is absent and an individual gets two defective recessive genes (nn).

1) What **phenotype** would an individual with the genotype NN be? _____

2) What **phenotype** would an individual with genotype Nn be? _____

3) A man heterozygous for albinism has 4 children with an albino woman. What phenotypes and genotypes would you expect?

   What is the man's genotype: _____

   What gene will be present in each of the man's different gametes: _____ , _____.

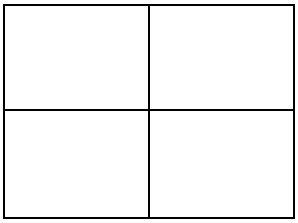
   What will be the woman's genotype: _____

   What will be present in each of the woman's gametes: _____, _____.

   In the punnett square below, place the two possible gametes for the woman at the top of each column and the two possible gametes for the man at the side of each row.

   Determine the genotypes of the offspring.

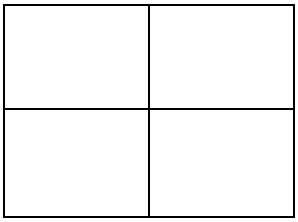
   Construct a Punnett square:



4) A couple that both have normal (dominant trait) skin color have an albino (recessive trait) child and 2 normal children. What are the parents' genotypes?

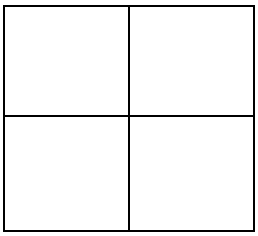
Mother's Genotype: _____ Father's Genotype: _____

Draw the punnett square showing all the possible genotypes for their children.



5) What are the chances of a couple both hybrid (heterozygous) for albinism having a 'normal' child?

Construct a Punnett square:



How many different possible children could be produced (how many squares)? _____

How many children contain at least one dominant gene in their genotype? _____

Divide the # showing the dominant trait by the total # of possible offspring, express as a percentage:

_____

Question Set B.

Cystic Fibrosis is a disabling genetic disease that used to always cause death by the time a child was a teenager. Its cause is a recessive gene (**g**) that produces abnormal glycoproteins that interfere with salt metabolism. The sweat of the body is high in sodium chloride (table salt), but more important, the mucus secreted is abnormally thick and clogs passageways in the lungs, liver and pancreas. This leads to pneumonia, and liver function is impaired so bile production is reduced and fat digestion is not normal. Digestive enzymes from the pancreas are interfered with by fibrous growths.

- Is Cystic Fibrosis a dominant or a recessive trait? _____
-
- What is the genotype of an individual with Cystic Fibrosis?_____
-
- What are the chances of two parents that are heterozygous (carriers) of the disease having a normal child?

Construct a Punnett square that shows the cross of the two parents described:

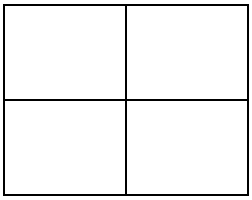
Probability of having 1 normal (CF free) child: _____ %

Question Set C.

Most genes that cause abnormal conditions (such as albinism) are recessive. An exception to this is **Chondrodystrophic dwarfism**. It is caused by the presence of a dominant gene (**C**). Most people are homozygous recessive (**cc**) and have normal height. People that are hybrid or homozygous dominant for the gene are dwarfs with shortened arms and legs, which are usually also very bowed, but their torsos are of normal size. The dominant gene causes premature closure of the growth plates in the arm and leg long bones.

6) What are the chances of two 'normal' people having a Chondrodystrophic dwarf child? _____

Construct a Punnett square:



Di-Hybrid Crosses:

1. Hearing (D) is dominant to deafness (d), and having a normal metabolism (P) is dominant to having pku (p). A married couple, Barbie and Ken, have these phenotypes and genotypes: Barbie is ddPP (deaf but normal metabolism), and Ken is Ddpp (he can hear but has pku). They ask their genetic counselor, "What are our chances of having a 'normal' child? (can hear and has normal metabolism)?

What is the genotype of 'Barbie'? _____

What will be the possible gamete combinations for 'Barbie'

1st of each gene pair _____

1st & 2nd _____

2nd & 1st _____

2nd of each pair _____

What is the genotype of 'Ken'? _____

What will be the possible gamete combinations for 'Ken'

1st of each gene pair _____

1st & 2nd _____

2nd & 1st _____

2nd of each pair _____

Draw a punnett square showing the possible offspring from Barbie and Ken.

Place the gametes from Barbie across the top (four columns), and the gametes from Ken along the side (rows).

XXXX	XXXX	**BARBIE**			
XXXX	XXXX				
K					
E					
N					

How many of Barbie & Ken's offspring will be:

Deaf but with normal metabolism _____

Deaf but with PKU _____

Hearing but with normal metabolism _____

Hearing but with PKU _____

2. In people nearsightedness (N) is dominant to normal vision (n), and polydactyly (having an 'extra' finger or toe) is dominant to 5-fingered hands. A man with normal vision and sho is heterozygous for polydactyly marries a heterozygous nearsighted woman with 'normal' hands.

What are the couple's chances (%) of having a child with normal vision and normal hands?

Draw a punnett square showing the results of this cross.

XXXX	XXXX	**FEMALE**			
XXXX	XXXX				
M A L E					